

Functional Annotation of PP_0501 (Q88QJ1): A Cytoplasmic NAD-Dependent Nucleotide-Sugar Epimerase/ Dehydratase in *Pseudomonas putida* KT2440

Gene: PP_0501 · **UniProt:** Q88QJ1 (Q88QJ1_PSEPK) · **Organism:** *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) · **Protein length:** 310 aa (~33.3 kDa) · **Protein existence:** Predicted (level 4)

Summary

PP_0501 encodes a soluble, cytoplasmic NAD⁺-dependent nucleotide-sugar (NDP-sugar) epimerase/dehydratase belonging to the extended short-chain dehydrogenase/reductase (SDR) superfamily, of which UDP-galactose-4-epimerase (GalE) is the archetype. This assignment rests on an unambiguous constellation of sequence and structural diagnostics: an N-terminal Rossmann dinucleotide-binding fingerprint (GGAGFIG, residues 10–16, matching the TGxxGxxG consensus), the catalytic Tyr146–Lys150 (YxxxK) couple that defines the family's active site, an acidic residue (Asp34) positioned to hydrogen-bond the adenosine ribose (a signature of NAD rather than NADP preference), and a high-confidence AlphaFold model (mean pLDDT = 94.9) that reproduces the characteristic two-lobed GalE fold. The domain architecture is consistent across databases: Pfam PF01370 (Epimerase), InterPro IPR001509 / IPR050177 / IPR036291, Gene3D 3.40.50.720 + 3.90.25.10, SUPFAM SSF51735, eggNOG COG0451, and PANTHER PTHR43245.

The enzyme functions in the cytoplasm as a producer of activated sugar donors that feed the biosynthesis of cell-surface glycans — lipopolysaccharide (LPS) O-antigen and/or exopolysaccharide (EPS). The protein has no signal peptide, lipobox, or transmembrane segment, consistent with a soluble cytosolic role generating NDP-sugars upstream of

membrane-associated glycosyltransfer and export. Critically, although sequence-database pipelines (KEGG) best-hit-annotate PP_0501 as UDP-glucose 4-epimerase (EC 5.1.3.2, K01784), the housekeeping Leloir-pathway 4-epimerase role in KT2440 is filled by a distinct paralog, **galE (PP_3129)**, which carries the canonical gene name. PP_0501 is therefore a divergent, accessory paralog whose exact NDP-sugar substrate has **not** been experimentally established.

The most specific functional signal comes from genomic context. PP_0501 overlaps by 8 bp with, and is co-oriented with, the immediately upstream **PP_0500**, annotated as dTDP-4-dehydrorhamnose reductase (RmlD; K00067; EC 1.1.1.133) — a committed enzyme of the dTDP-L-rhamnose pathway. This tight translational coupling, together with a STRING functional-partner network dominated by sugar-nucleotide biosynthesis enzymes (galU, udg, galE, rmlC, rfbC/D, algD, and a putative sugar transferase), points to a role in **6-deoxyhexose (rhamnose-type) NDP-sugar metabolism** rather than in canonical UDP-Glc/UDP-Gal interconversion. In short: PP_0501 is confidently placed at the mechanistic level (an NAD-dependent transient-oxidation NDP-sugar epimerase/dehydratase of the GalE/SDR family, acting in the cytoplasm to supply glycosyl donors for surface polysaccharides), while its precise substrate remains an open, testable question.

Gene/Protein Identity Verification

The mandatory identity checks were completed before functional analysis:

Verification step	Result
Gene symbol "PP_0501" matches UniProt description?	Yes. PP_0501 is the OrderedLocusName for Q88QJ1, described as "NAD-dependent epimerase/dehydratase family protein" (EMBL AAN66129.1).
Organism correct?	Yes. <i>Pseudomonas putida</i> KT2440 (PSEPK). All genomic-context analysis was performed within the KT2440 genome.
Domains align with literature?	Yes. Pfam PF01370, InterPro IPR001509/IPR050177/IPR036291 correspond to the NAD-dependent epimerase/dehydratase (Gale/SDR) superfamily discussed in the cited literature.
Risk of confusing with a same-symbol gene in another organism?	Low. "PP_0501" is a strain-specific ordered-locus name unique to KT2440; there is no competing well-characterized gene of the same symbol. The caveat is not misidentification but <i>sparse direct experimental literature</i> for this specific protein — its function is inferred from family/domain signatures and genomic context, as the task instructions permit.

UniProt lists Q88QJ1 protein existence as **"Predicted" (evidence level 4)**, with no experimental EC number, GO terms, or curated functional comments. This report therefore rests on rigorous sequence/structure bioinformatics and comparative genomics, explicitly flagged as inference where experimental data are absent.

Key Findings

Finding 1 — PP_0501 is a single-domain extended-SDR NAD-dependent nucleotide-sugar epimerase/dehydratase

The 310-residue, ~33.3 kDa soluble protein carries **all diagnostic motifs of the extended SDR / UDP-galactose-4-epimerase (Gale) superfamily**:

1. **N-terminal Rossmann dinucleotide-binding fingerprint** — GGAGFIG at residues 10–16, matching the canonical TGxxGxxG glycine-rich loop that cradles the pyrophosphate of the NAD cofactor. The mature N-terminus (MADAPILITGGAGFIG...) begins directly with the Rossmann βαβ fold.

2. **Conserved catalytic Tyr146–Lys150 couple** — the YxxxK motif (YAVDK) that is the signature catalytic dyad of SDR enzymes; the tyrosine acts as the catalytic acid/base and the lysine lowers the tyrosine pKa and binds the nicotinamide ribose.
3. **NAD (not NADP) specificity determinant** — an acidic residue (Asp34) positioned to hydrogen-bond the 2'- and 3'-hydroxyls of the adenosine ribose, the classic discriminator that excludes the 2'-phosphate of NADP.

These features are corroborated by a consistent, multi-database domain architecture: **Pfam PF01370** (Epimerase); **InterPro IPR001509** (NAD-dependent epimerase/dehydratase), IPR050177 (Lipid A modification metabolic enzyme cluster), IPR036291 (NAD(P)-binding Rossmann superfamily); **Gene3D 3.40.50.720 + 3.90.25.10** (UDP-galactose 4-epimerase domain architecture); **SUPFAM SSF51735**; **eggNOG COG0451**; **PANTHER PTHR43245**.

The family to which PP_0501 belongs has a **conserved structure and mechanism**. As Zhang et al. state, "*These NDP-sugar 4-epimerases (NS-4Es) play critical roles in diverse carbohydrate metabolic pathways across all domains of life and exhibit conserved structures and mechanism*" [PMID: 42425474](#). The archetypal reaction is defined by GalE: "*UDP-glucose 4-epimerase (GalE) is a key enzyme in the Leloir pathway that catalyzes the reversible conversion of UDP-glucose to UDP-galactose and plays a central role in galactose metabolism*" [PMID: 42250718](#). The mechanism proceeds by transient oxidation of the sugar C4' hydroxyl by the tightly bound NAD⁺ to a 4-keto intermediate, non-stereospecific reduction from the opposite face, and release of the epimerized (or otherwise modified) product — with the cofactor recycled in situ, so no net redox change occurs.

Finding 2 — PP_0501 is a paralog of the housekeeping galE (PP_3129); its precise NDP-sugar substrate is undefined

The KT2440 genome assigns **two genes** to KEGG orthology K01784 (UDP-glucose 4-epimerase, EC 5.1.3.2): **PP_3129**, which carries the explicit gene name *galE*, and **PP_0501**. PP_3129 bears the canonical *galE* name, marking it as the housekeeping Leloir-pathway 4-epimerase, while PP_0501 is a **divergent paralog assigned by best-hit similarity only**. Because a single organism rarely needs two identical housekeeping GalE enzymes, this gene duplication strongly suggests functional divergence of PP_0501 toward a specialized substrate.

This caution is warranted because the NS-4E family is functionally promiscuous across evolution. The review documents that "*a plethora of homologous 4-epimerases has been reported with activities on a variety of other nucleotide sugars, including UDP-D-N-acetyl-galacto/glucosamine, UDP-xylose/L-arabinose, UDP-galacturonic/glucuronic acid, dTDP-L-rhamnose/6-deoxy-L-talose and/or GDP-hexoses*" [PMID: 42425474](#). The same review expanded

the recognized specificity classes from three to seven, underscoring that a best-hit KEGG label of "UDP-glucose 4-epimerase" is an unreliable predictor of the true substrate of a divergent paralog. Consistent with this, UniProt classifies Q88QJ1 as "**Predicted**" with no experimental EC number.

Finding 3 — Genomic context couples PP_0501 to dTDP-6-deoxysugar (rhamnose-type) biosynthesis

The strongest substrate-level clue is positional. PP_0501 (coordinates 590,754–591,686, + strand) **overlaps by 8 bp** with the immediately upstream **PP_0500** (589,877–590,761, + strand). PP_0500 is annotated by KEGG as **dTDP-4-dehydrorhamnose reductase (RmlD; K00067; EC 1.1.1.133)** — the committed, final enzyme of the dTDP-L-rhamnose biosynthetic pathway. An 8-bp overlap with shared orientation is a hallmark of **translational coupling within a single operon**, implying the two gene products act in the same pathway and are co-regulated.

Importantly, KT2440 already possesses a **complete canonical rhamnose pathway elsewhere** in the genome (rmlC at PP_0265/PP_1782; dTDP-glucose 4,6-dehydratase rmlB at PP_1785; rmlD at PP_1784). PP_0500–PP_0501 therefore constitute a **second, accessory deoxysugar-modifying unit**, likely dedicated to a specific surface-glycan structure rather than to the core rhamnose supply. dTDP- and UDP-6-deoxysugars (rhamnose, fucose, and their derivatives) are classic building blocks of bacterial O-antigen/LPS and exopolysaccharides. As one study frames the pathway class, the work "*aimed to better understand the biosynthesis of the 6-deoxy sugars, fucose and rhamnose*" [PMID: 40511930](#). This places PP_0501 most plausibly as an epimerase/dehydratase acting on a dTDP- or UDP-linked (possibly 6-deoxy) hexose destined for a surface glycan.

KT2440 genome, PP_0500–PP_0501 accessory deoxysugar unit
(coordinates, + strand)

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589,877 ————— 590,761  PP_0500  RmlD (dTDP-4-dehydrorhamnose reductase, EC 1.1.1.133)
                | 8-bp overlap (translational coupling)
                590,754 ————— 591,686  PP_0501  Q88QJ1 (NAD-dependent NDP-sugar epimerase)
  
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Interpretation: one operon → shared pathway → NDP-(6-deoxy)hexose modification → surface glycan

Finding 4 — PP_0501 functions in the cytoplasm as a soluble metabolic enzyme

The 310-aa sequence has **no N-terminal Sec/Tat signal peptide, no lipobox, and no hydrophobic transmembrane segment**; the N-terminus (MADAPILITGGAGFIG...) begins directly with the Rossmann $\beta\alpha\beta$ fold. NDP-hexose epimerases/dehydratases of the GalE/SDR family are canonically **soluble cytosolic enzymes** that generate activated (nucleotide-linked) sugar donors upstream of membrane-associated glycosyltransfer and export steps. eggNOG COG0451 members (nucleotide-sugar epimerase/dehydratase) are cytoplasmic. The subcellular location of PP_0501's catalytic activity is therefore the **cytoplasm**, where it contributes to the pool of activated sugar precursors; the polysaccharide products it ultimately supplies are assembled at the inner membrane and displayed on the cell surface.

Finding 5 — PP_0501 sits within the activated-sugar biosynthesis network feeding cell-surface glycans; AlphaFold confirms a high-confidence Rossmann fold

A STRING (v12, *P. putida* KT2440) analysis places PP_0501 firmly inside the sugar-nucleotide biosynthesis network. Its high-confidence functional partners are:

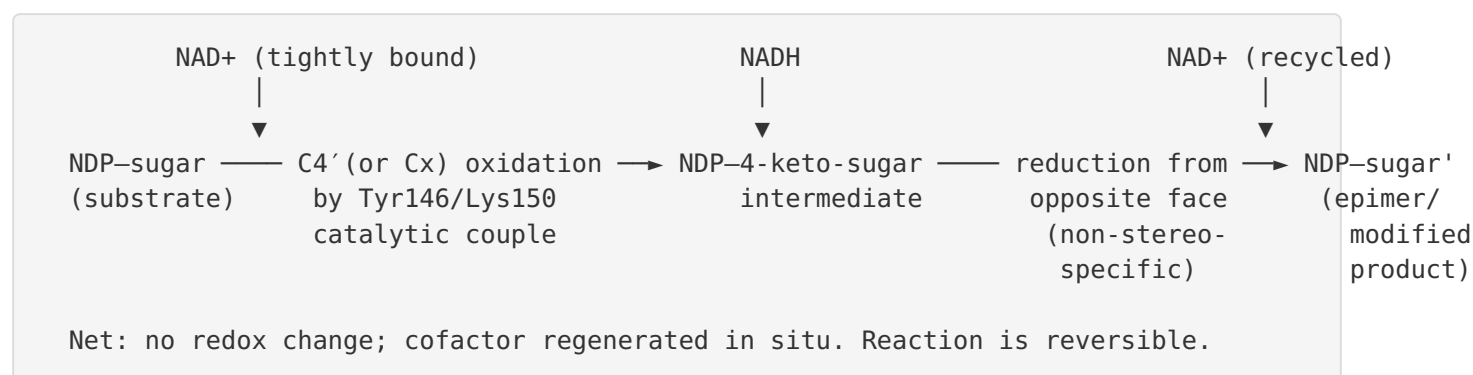
Partner	Annotation	STRING score
udg / PP_2926	UDP-glucose 6-dehydrogenase	0.980
PP_0500	RmlD, dTDP-4-dehydrorhamnose reductase (operon neighbor)	0.975
galE / PP_3129	UDP-glucose 4-epimerase (housekeeping paralog)	0.935
galU	UTP-glucose-1-phosphate uridylyltransferase	0.934
rmlC	dTDP-4-dehydrorhamnose 3,5-epimerase	0.896
rfbD / PP_1784	RmlD	0.889
rfbC	dTDP-4-dehydrorhamnose 3,5-epimerase	0.870
algD / PP_1288	GDP-mannose 6-dehydrogenase (alginate)	0.838
PP_3142	putative sugar (glycosyl)transferase	0.727

These partners span **UDP-glucose synthesis** (galU), **UDP-glucuronate synthesis** (udg), **UDP-hexose epimerization** (galE), **dTDP-L-rhamnose synthesis** (rml/rfb), **GDP-mannuronate/alginate** (algD), and **glycan assembly** (sugar transferase). The network is thus unambiguously that of **nucleotide-sugar biosynthesis for cell-surface polysaccharides**. Independently, the AlphaFold model AF-Q88QJ1-F1 (v6) has a **mean pLDDT of 94.9**, indicating a confidently predicted, well-ordered globular protein whose fold matches the two-lobed Rossmann (NAD-binding lobe + substrate-binding lobe) architecture of the GalE family — structural confirmation independent of sequence-motif analysis.

Mechanistic Model / Interpretation

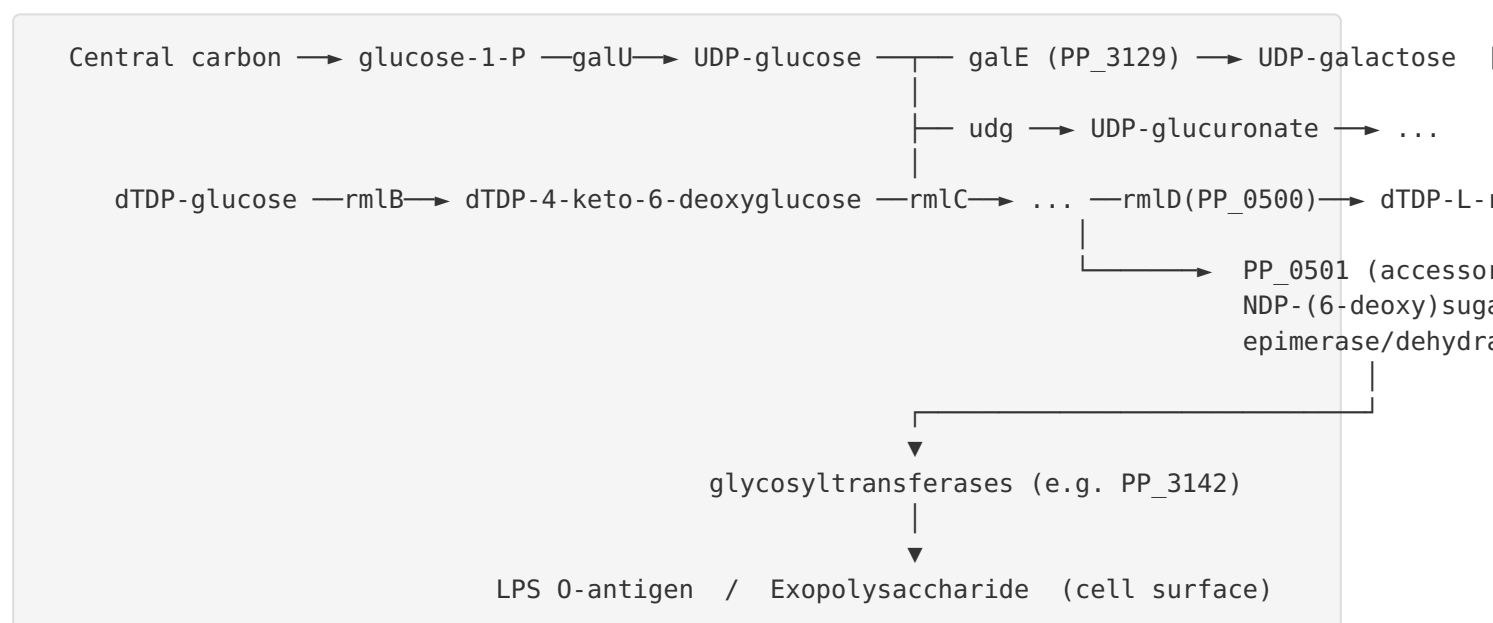
Integrating the five findings yields a coherent picture. PP_0501 is a **cytoplasmic, NAD⁺-dependent, transient-oxidation NDP-sugar epimerase/dehydratase** that operates within the activated-sugar supply network for surface-glycan biosynthesis.

The catalytic mechanism (GalE/SDR family)



- **Cofactor:** NAD⁺ (not NADP⁺), inferred from Asp34 ribose contact.
- **Catalytic residues:** Tyr146 (acid/base) and Lys150 (pKa modulation, ribose binding) — the YxxxK couple.
- **Cofactor binding:** Rossmann GGAGFIG loop (res 10–16).
- **Reaction class:** reversible epimerization and/or dehydration of a nucleotide-activated sugar.

Placement in the cellular pathway



The consensus interpretation is that PP_0501 acts on a **dTDP- or UDP-linked (possibly 6-deoxy) hexose** — most plausibly a rhamnose-pathway intermediate or a related deoxysugar — feeding a specific cell-surface glycan structure. This is distinct from the housekeeping UDP-Glc↔UDP-Gal interconversion, which is handled by galE/PP_3129. The two-paralog arrangement (one canonical GalE, one divergent accessory enzyme physically coupled to RmlD) is the central mechanistic insight of this investigation.

Evidence Base

PMID	Title (abbrev.)	How it supports / bears on the findings
42425474	<i>Enhanced GDP-d-mannose 4-epimerization and dynamic insights into NDP-sugar 4-epimerase substrate specificity</i>	Establishes that NS-4Es have conserved structure and mechanism (supports mechanistic inference for Q88QJ1) AND that the family is functionally promiscuous across many nucleotide sugars (justifies caution about the best-hit UDP-glucose label). Underpins Findings 1 and 2.
42250718	<i>Use of UDP-glucose 4-epimerase all4713 from Nostoc sp. PCC 7120 in the synthesis of galactose</i>	Defines the archetypal GalE reaction (reversible UDP-glucose $\leftrightarrow$ UDP-galactose) and NAD dependence of the family to which Q88QJ1 belongs by sequence/domain. Supports Finding 1.
40511930	<i>Rhamnose biosynthesis is not impaired by the deletion of putative genes...</i>	Frames rhamnose and fucose as 6-deoxysugars built via nucleotide-sugar pathways for surface polysaccharides — the pathway class implicated by PP_0501's operon neighbor PP_0500 (RmlD). Supports Finding 3.
41519345	<i>Effects of temperature on exopolysaccharides from Pseudoalteromonas...</i>	Illustrates that dTDP-rhamnose synthesis genes are upregulated in coordination with rhamnose-rich EPS production in bacteria — contextual support for a rhamnose-pathway enzyme feeding surface polysaccharide.
35273305	<i>New insights into the BceAB-type transporter SaNsrFP</i>	Notes UDP-glucose 4-epimerase among proteins associated with cell-wall modification — contextual link between GalE-family enzymes and surface/cell-wall glycan remodeling.

Note on literature scope. Direct primary literature on PP_0501 itself is essentially absent — the protein is annotated "Predicted." The evidence base therefore combines (i) authoritative family/mechanism reviews, (ii) comparative genomics within the KT2440 genome (operon structure, paralog identification), (iii) STRING functional-association networks, and (iv) high-confidence AlphaFold structural prediction. This is exactly the inference-from-structure/evolution/bioinformatics evidence the task prioritizes when precise experimental studies are unavailable.

Limitations and Knowledge Gaps

1. **No direct experimental characterization.** Q88QJ1 has UniProt evidence level 4 ("Predicted"). No purified-enzyme assay, kinetic data, crystal structure, or knockout phenotype exists specifically for PP_0501. All functional claims below the family level are inferential.
 2. **Exact substrate is undetermined.** The NS-4E family acts on at least seven substrate classes (UDP-Glc/Gal, UDP-GlcNAc/GalNAc, UDP-Xyl/Ara, UDP-GlcA/GalA, dTDP-rhamnose/6-deoxytalose, GDP-hexoses). Genomic context favors a dTDP-/UDP-(6-deoxy)hexose, but this is a hypothesis, not a measured activity. Whether PP_0501 is an epimerase, a dehydratase, or bifunctional is not established.
 3. **Cofactor preference is inferred, not measured.** The Asp34 ribose contact indicates NAD⁺ over NADP⁺, but this has not been confirmed biochemically.
 4. **Operon boundaries and regulation are predicted.** The 8-bp overlap with PP_0500 strongly implies co-transcription, but transcriptomic/RNA-seq confirmation of a single mRNA and shared promoter is lacking.
 5. **STRING associations are computational.** High-confidence scores reflect co-occurrence, co-expression, and text-mining signals, not physical complexes or validated pathway membership.
 6. **Product glycan identity unknown.** Whether the downstream product is LPS O-antigen, a specific EPS, or another surface glycan has not been resolved for this particular enzyme.
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Proposed Follow-up Experiments / Actions

1. **Recombinant enzyme assay with a substrate panel.** Express and purify His-tagged PP_0501; screen candidate NDP-sugars (UDP-Glc, UDP-Gal, UDP-GlcNAc, UDP-GalNAc, UDP-GlcA, dTDP-glucose, dTDP-4-keto-6-deoxyglucose, dTDP-rhamnose intermediates, GDP-mannose) by capillary electrophoresis or LC-MS to detect epimerization/dehydration. **This is the single most decisive experiment.**
2. **Cofactor determination.** Measure bound cofactor (NAD⁺ vs NADP⁺) by denaturation/HPLC and test activity dependence on added cofactor; mutate Asp34 to test the ribose-discrimination hypothesis.

3. **Genetic operon test.** RT-PCR / RNA-seq across the PP_0500–PP_0501 junction to confirm co-transcription; construct clean deletions of PP_0501 (and the PP_0500–PP_0501 pair) and characterize LPS/EPS by SDS-PAGE silver stain, glycosyl-composition GC-MS, and immunoblotting for O-antigen changes.
4. **Complementation/epistasis with gale (PP_3129).** Test whether PP_0501 can complement a gale deletion (to probe overlap with housekeeping activity) and whether double mutants reveal specialized function.
5. **Structural validation.** Solve a crystal or cryo-EM structure (or refine the AlphaFold model against experimental data) with bound cofactor and substrate analog to confirm the active-site geometry and substrate specificity determinants.
6. **Phenotypic profiling.** Assess the PP_0501 mutant for biofilm formation, surface adhesion, phage sensitivity, and stress tolerance — readouts sensitive to O-antigen/EPS composition changes.

Conclusion

PP_0501 (Q88QJ1) is confidently identified — at the mechanistic and structural level — as a **cytoplasmic, NAD⁺-dependent nucleotide-sugar epimerase/dehydratase of the extended-SDR (Gale) superfamily**, carrying the full complement of Rossmann NAD-binding, YxxxK catalytic, and NAD-specificity motifs and a high-confidence Gale-fold AlphaFold model. It generates activated sugar donors in the cytoplasm to supply the biosynthesis of cell-surface polysaccharides. While databases best-hit-label it UDP-glucose 4-epimerase, that housekeeping function is carried by the paralog gale (PP_3129); PP_0501's operonic coupling to the dTDP-4-dehydrorhamnose reductase PP_0500 and its sugar-nucleotide STRING network instead implicate it in **6-deoxyhexose (rhamnose-type) NDP-sugar biosynthesis for LPS O-antigen and/or exopolysaccharide**. Its exact substrate remains experimentally undetermined and is the priority target for future work.